

● EASY

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

01

10 questions · ~15 min

Q1

ADVANCED MATH

$$x = 3$$

$$y = 15 - x^2$$

A solution to the given system of equations is x, y . What is the value of xy ?

A. 432

B. 54

C. 45

D. 18

Q2

GRID-IN

ADVANCED MATH

$$5x = 45$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

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.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

$$x = 49$$

$$y = \sqrt{x} + 9$$

The graphs of the given equations intersect at the point x, y in the xy -plane. What is the value of y ?

- A. 16
- B. 40
- C. 81
- D. 130

Q4

ADVANCED MATH

$$3x^2 - 15x + 18 = 0$$

How many distinct real solutions are there to the given equation?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Q5

ADVANCED MATH

$$6x - 9y > 12$$

Which of the following inequalities is equivalent to the inequality above?

- A. $x - y > 2$
- B. $2x - 3y > 4$
- C. $3x - 2y > 4$
- D. $3y - 2x > 2$

Q6

ADVANCED MATH

$$p + 61 = 65$$

Which value is a solution to the given equation?

- A. $\frac{65}{61}$
- B. 4
- C. 126
- D. 130

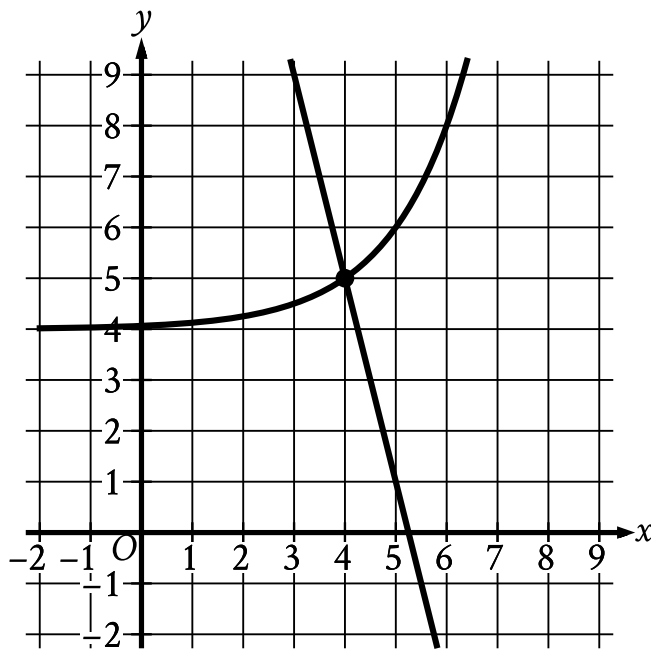
Which of the following is a solution to the equation $2x^2 - 4 = x^2$?

A. 1

B. 2

C. 3

D. 4



- For the curve in the system:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to the approximate point (0 comma 4.063).
 - In quadrant 1:
 - The curve trends up gradually to point (4 comma 5).
 - The curve then trends up sharply.
 - The curve passes through the following points:
 - (3 comma nine halves)
 - (4 comma 5)
 - (5 comma 6)
- For the line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (3 comma 9)
 - (4 comma 5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

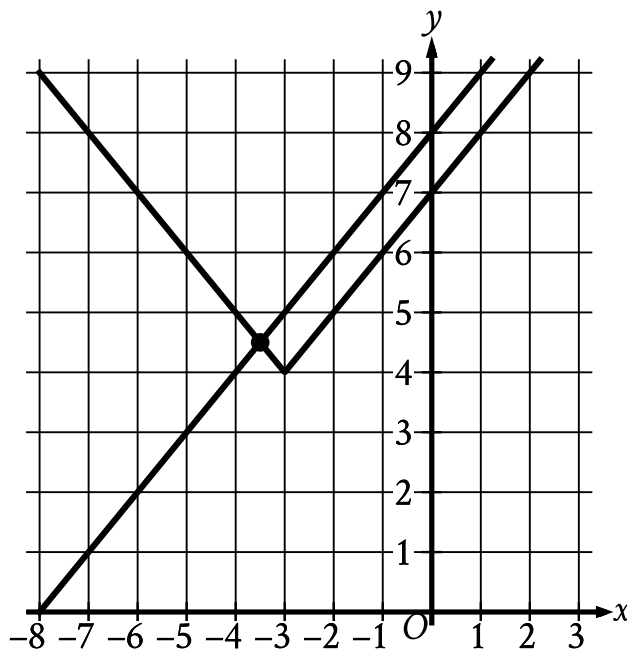
- A. 0,0
- B. 0,4
- C. 4,5
- D. 5,0

Q9

ADVANCED MATH

If $(x+5)^2 = 4$, which of the following is a possible value of x ?

- A. 1
- B. -1
- C. -2
- D. -3



- For the absolute value function in the system:
 - Moving from left to right:
 - The function slants sharply down to the point (negative 3 comma 4).
 - The function then slants sharply up.
 - The function passes through the following points:
 - (negative seven halves comma nine halves)
 - (negative 3 comma 4)
 - (negative five halves comma nine halves)
- For the linear function in the system:
 - The function slants sharply up from left to right.
 - The function passes through the following points:
 - (negative seven halves comma nine halves)
 - (0 comma 8)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. 0,8

B. $\frac{7}{2}, \frac{9}{2}$

C. $-\frac{7}{2}, \frac{9}{2}$

D. -3,4